

Q16) Write a C program that can perform a letter frequency attack on any monoalphabetic substitution cipher without human intervention. Your software should produce possible plaintexts in rough order of likelihood. It would be good if your user interface allowed the user to specify “give me the top 10 possible plaintexts.”

Aim: To write a C program to perform a **letter frequency attack** on a monoalphabetic substitution cipher and generate possible plaintexts based on English letter frequency analysis.

Algorithm:

1. Start the program.
2. Read the ciphertext from the user.
3. Count the frequency of each alphabet letter.
4. Sort the letters according to their frequencies.
5. Compare them with the English frequency order:
ETAOINSHRDLCMWFGYPBVKJXQZ
6. Replace the most frequent ciphertext letter with **E**, second with **T**, third with **A**, and so on.
7. Generate the corresponding plaintext.
8. Display the probable plaintext.
9. Repeat the mapping (if required) to generate additional candidate plaintexts.
10. Stop the program.

Result:

Thus, the C program to perform a **letter frequency attack** on a **monoalphabetic substitution cipher** was successfully executed, and a probable plaintext was generated using English letter frequency analysis.

Code:

```
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(globals)
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#include <stdio.h>
#include <string.h>
#include <ctype.h>
int main()
{
    char cipher[1000];
    int freq[26]={0};
    int i,j,max,temp;
    char letters[26];
    char english[]="ETAOINSHRDLCKUMWFGYPBVKJXQZ";
    printf("Enter Ciphertext:\n");
    fgets(cipher,sizeof(cipher),stdin);
    // Count frequency
    for(i=0; cipher[i]!='\0'; i++)
    {
        if(isalpha(cipher[i]))
            freq[toupper(cipher[i])-'A']++;
    }
    // Store alphabet
    for(i=0;i<26;i++)
        letters[i]='A'+i;
    // Sort frequencies
    for(i=0;i<25;i++)
    {
        for(j=i+1;j<26;j++)
        {
            if(freq[j]>freq[i])
            {
                temp=freq[i];
                freq[i]=freq[j];
                freq[j]=temp;
            }
        }
    }
    printf("Letter Frequencies:\n");
    for(i=0;i<26;i++)
        printf("%c : %d\n",letters[i],freq[i]);
    printf("\nProbable Plaintext:\n");
    for(i=0; cipher[i]!='\0'; i++)
    {
        if(isalpha(cipher[i]))
        {
            char ch=toupper(cipher[i]);
            for(j=0;j<26;j++)
            {
                if(ch==letters[j])
                {
                    printf("%c",english[j]);
                    break;
                }
            }
        }
        else
            printf("%c",cipher[i]);
    }
    return 0;
}
```

```
Untitled1 - Dev-C++ 5.11
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(globals)
Project Classes Debug [?] Untitled1

    char c=letters[i];
    letters[i]=letters[j];
    letters[j]=c;
}
}
printf("\nLetter Frequencies\n");
for(i=0;i<26;i++)
    printf("%c : %d\n",letters[i],freq[i]);
printf("\nProbable Plaintext:\n");
for(i=0; cipher[i]!='\0'; i++)
{
    if(isalpha(cipher[i]))
    {
        char ch=toupper(cipher[i]);
        for(j=0;j<26;j++)
        {
            if(ch==letters[j])
            {
                printf("%c",english[j]);
                break;
            }
        }
    }
    else
        printf("%c",cipher[i]);
}
return 0;
}
```

Output:

```
Select C:\Users\SAIL\Desktop\lab16.exe
Enter Ciphertext:
WKL V LV D VHFUHW

Letter Frequencies
V : 3
L : 2
H : 2
W : 2
F : 1
U : 1
D : 1
K : 1
I : 0
J : 0
A : 0
E : 0
M : 0
N : 0
O : 0
P : 0
Q : 0
R : 0
S : 0
T : 0
B : 0
G : 0
C : 0
X : 0
Y : 0
Z : 0

Probable Plaintext:
OHT E TE S EAINAO

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Process exited after 3.169 seconds with return value 0
Press any key to continue . . .
```